



The Role of Lower Limb Joints in Classifying Gait Patterns in Children with Cerebral Palsy Using Machine Learning

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Abstract

This study formulated the classification of four gait patterns in children with cerebral palsy, as defined by the Rodda system, as a supervised machine learning problem. We examined how hip, knee, and ankle kinematics contribute to distinguishing these patterns and how combinations of joints affect classifier performance. Sagittal-plane joint-angle trajectories were evaluated as single-joint, two-joint, and three-joint feature sets. Four classifiers were trained: k-nearest neighbors (KNN), support vector machines (SVM), random forests (RF), and logistic regression (LR). A single joint did not adequately distinguish the four groups; the knee was the most informative individual joint. Combining joints, particularly the knee and ankle, substantially improved classification. The best results were obtained with all three joints, consistent with the multijoint nature of the gait patterns. Joint-specific analysis can thus improve classification while providing insight into the biomechanical contribution of each joint.

Keywords

Cerebral palsy · Gait analysis · Rodda classification · Machine learning

1 Introduction

Cerebral palsy (CP) is the most common movement disorder of childhood. It comprises permanent neurological disorders that affect the development of movement and can cause motor impairments and activity limitations. The underlying brain disturbance may arise before, during, or after birth. Approximately 2 to 3 children per 1,000 live births have CP [1, 2].

Classification systems for movement disorders give clinicians a shared language and can support treatment planning [3]. A widely used approach in CP is the Rodda classification, which describes four sagittal-plane gait patterns: true equinus, jump gait, apparent equinus, and crouch gait [4, 5]. Experienced assessors have shown acceptable agreement when applying this framework, and it can help describe gait changes associated with growth and orthopedic intervention [6]. Nevertheless, the classification relies mainly on visual interpretation of hip, knee, and ankle motion. It therefore depends on the assessor's experience and may vary between assessors.

Supervised machine learning methods have increasingly been applied to biomechanics and gait analysis. By learning from labeled examples, they can identify discriminative patterns in kinematic data and assign observations to the predefined clinical classes. For example, Zhang and colleagues used a support vector machine to distinguish fatigued from normal walking, reporting 96% accuracy [7]. Armand and colleagues used fuzzy decision trees to classify toe-walking patterns, reporting 81% accuracy [8].

Several studies have classified gait in CP or distinguished children with CP from typically developing peers. Kamruzzaman and Begg used a support vector machine with spatiotemporal inputs, including step length and cadence, to distinguish CP gait with 83.3% accuracy [9]. Figueiredo and colleagues reported strong support vector machine performance in gait-pattern recognition, particularly for CP. Zhang and Ma compared seven learning methods for sagittal gait-pattern classification and reported 93.5% accuracy with a neural network [11].

Prior work supports the use of machine learning for gait classification, but it has focused primarily on overall accuracy. The relative contribution of individual joints to a multiclass decision has received less systematic attention. In particular, it remains unclear whether the hip, knee, or ankle best distinguishes the sagittal-plane patterns and whether specific joint combinations provide additional information.

We therefore modeled the Rodda gait patterns as a supervised four-class problem and compared conventional machine learning methods across all single-joint, paired-joint, and three-joint configurations. This joint-specific comparison was intended to clarify which kinematic inputs are most informative and to support the development of objective tools for clinical gait analysis.

2 Methods

2.1 Participants and kinematic data acquisition

The dataset comprised children aged 5–13 years with spastic diplegic CP who attended the Javad Mowafaghian Rehabilitation Research Center for clinical gait analysis. The children had no history of orthopedic or neurosurgical treatment. Each limb was assigned to true equinus, jump gait, apparent equinus, or crouch gait on the basis of its sagittal-plane pattern following Rodda and colleagues [4, 5]. The two limbs of one child could receive different labels.

Participants walked barefoot at a self-selected speed along a 10 m laboratory walkway. All walked independently without a walker or cane for at least ten consecutive gait cycles. Three-dimensional kinematics were recorded at 120 Hz with a ten-camera Vicon motion-capture system. Reflective markers were placed on the pelvis and lower limbs following the Plug-in Gait protocol; the original acquisition description specifies 16 markers per limb. The acquired joint kinematics covered all three anatomical planes.

Only sagittal-plane ankle, knee, and hip angles were retained for this analysis. Each limb was represented by a 300-element vector consisting of 100 time-normalized samples for each joint, ordered ankle, knee, then hip. The final dataset contained 833 affected limbs: 294 true equinus, 152 apparent equinus, 201 jump gait, and 186 crouch gait. These limb-level observations were used to classify the four patterns.

2.2 Exploratory analysis of the data

Before model fitting, we examined the raw kinematic profiles and the extent to which the classes separated in feature space.

2.2.1 Mean joint-angle trajectories

For each Rodda group, the mean and variability of the hip, knee, and ankle angles were plotted over the gait cycle (Figure 1). The largest visual differences between groups appeared in knee motion during midstance. Crouch gait showed sustained knee flexion and was readily distinguishable from several other patterns. Ankle motion differed particularly between true equinus and crouch gait, reflecting their contrasting plantarflexion and dorsiflexion profiles. Hip-angle trajectories showed greater overlap across groups.

2.2.2 Principal component analysis

Principal component analysis (PCA) reduced each individual-joint feature set and the combined three-joint feature set to two components for visualization (Figure 2). The projected classes overlapped substantially; they were neither perfectly nor linearly separable in this two-dimensional representation.

Single-joint projections showed extensive overlap between classes. Combining all three joints produced somewhat more coherent clusters, although the classes still overlapped. This descriptive pattern suggests that information from multiple joints may improve classification, as gait patterns reflect coordinated motion rather than a single joint angle.

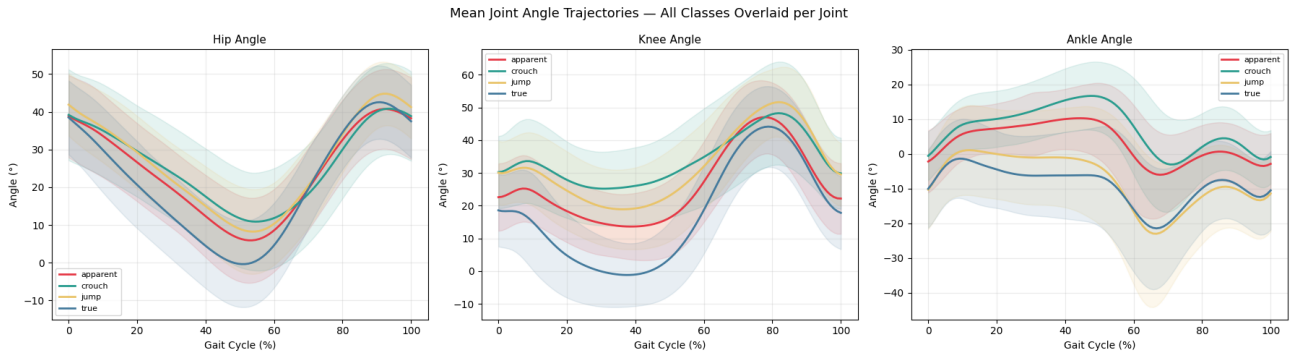


Figure 1. Mean sagittal-plane joint-angle trajectories and variability for the four Rodda gait groups.

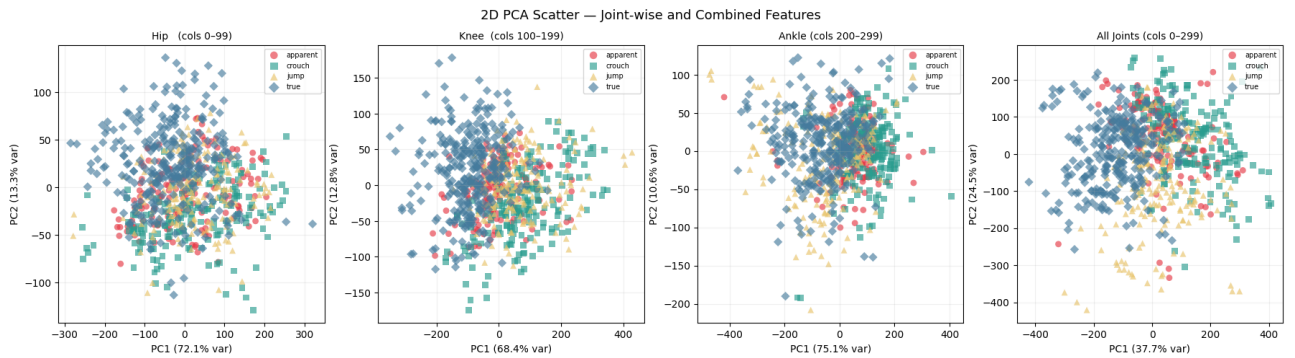


Figure 2. Two-dimensional PCA projections of individual-joint and combined-joint features

2.3 Data preprocessing

The raw dataset was preprocessed to improve model stability and allow comparisons across methods. Outlying observations were removed using z scores. Features were then standardized with StandardScaler to obtain zero mean and unit variance.

Data were divided by stratified sampling into training (80%) and test (20%) sets, preserving class proportions. Synthetic Minority Over-sampling Technique (SMOTE) was applied only to the training data to address class imbalance and avoid contamination of the test set.

2.4 Model fitting and hyperparameter tuning

Four multiclass classifiers were evaluated: KNN, SVM, RF, and LR. Hyperparameters were tuned with GridSearchCV using five-fold stratified cross-validation rather than default model settings.

For SVM, the search covered the regularization parameter C, linear and radial basis function kernels, and gamma. For RF, the number of trees, maximum tree depth, minimum samples per node, and number of features considered at each split were tuned. The best setting for each model was selected according to mean cross-validation accuracy.

Each selected model was refitted on the full training set and evaluated on the held-out test set. Accuracy, precision, recall, and

F1 score were reported to assess performance across the gait groups.

3 Results

3.1 Overall performance across joint configurations

Test-set classification accuracy for each joint configuration and model is shown in Table 1.

Table 1. Test-set accuracy by joint configuration and classifier.

Joints / model	SVM	RF	LR	KNN
Hip	0.5329	0.5509	0.4551	0.5329
Knee	0.6587	0.5749	0.5509	0.6647
Ankle	0.6228	0.5808	0.5329	0.6527
Hip + knee	0.7365	0.6467	0.5449	0.7665
Hip + ankle	0.7246	0.7186	0.5689	0.7904
Knee + ankle	0.7844	0.7545	0.6766	0.8144
All three joints	0.8204	0.7425	0.6527	0.8263

The corresponding F1 scores are reported in Table 2.

Table 2. F1 score by joint configuration and classifier.

Joints / model	SVM	RF	LR	KNN
Hip	0.5073	0.5200	0.4290	0.5188
Knee	0.6237	0.5322	0.5169	0.6412
Ankle	0.5940	0.5619	0.5113	0.6377
Hip + knee	0.7206	0.6104	0.5188	0.7502
Hip + ankle	0.7009	0.6910	0.5453	0.7794
Knee + ankle	0.7671	0.7278	0.6598	0.8030
All three joints	0.8081	0.7110	0.6347	0.8122

Performance depended strongly on the selected joints (Tables 1 and 2). With individual joints, accuracy ranged from approximately 0.45 to 0.66. The knee performed best: knee-only KNN achieved 0.6647 accuracy and an F1 score of 0.6412. Adding the ankle increased KNN accuracy to 0.8144 and F1 to 0.8030. All three joints produced the highest KNN accuracy (0.8263) and F1 score (0.8122); SVM achieved a similar accuracy of 0.8204 with this configuration.

LR generally yielded the lowest accuracy and F1 score among the four methods, consistent with limited linear separation of these kinematic patterns.

3.2 Evaluation metrics as joints are combined

Figures 3–6 show accuracy, precision, recall, and F1 score for KNN, SVM, RF, and LR, respectively, across the seven joint configurations. Together, these plots show how classification performance changed as joint information was combined.

Across models, the metrics generally improved as more joints were included. Performance was moderate with a single joint, increased with two-joint combinations, and was highest or near highest with all three joints.

KNN showed a largely consistent improvement across the configurations and benefited most from combining joints. LR improved less, consistent with its weaker overall performance. SVM and RF followed a broadly similar pattern to KNN, although their values were lower in several configurations.

These results suggest that discriminative information is distributed across the joints and that combined kinematic inputs improve the classification of Rodda gait patterns.

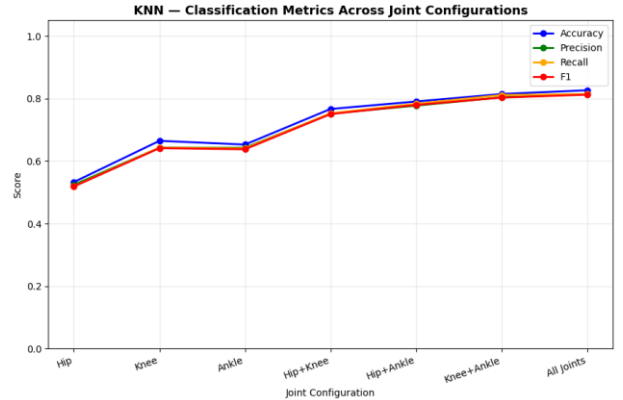


Figure 3. Accuracy, precision, recall, and F1 score for KNN across joint configurations.

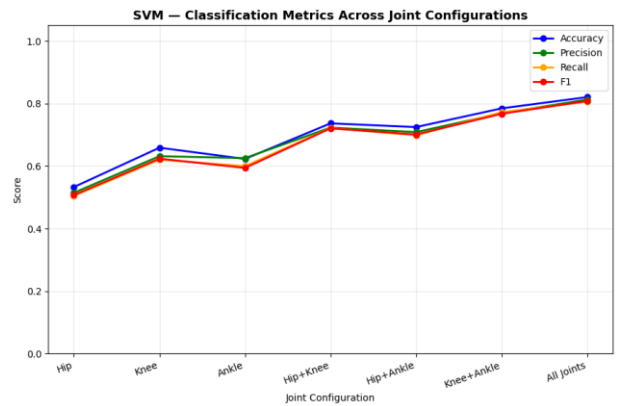


Figure 4. Accuracy, precision, recall, and F1 score for SVM across joint configurations.

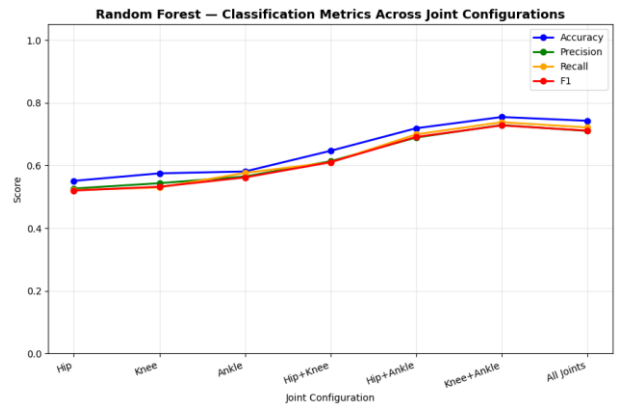


Figure 5. Accuracy, precision, recall, and F1 score for RF across joint configurations.

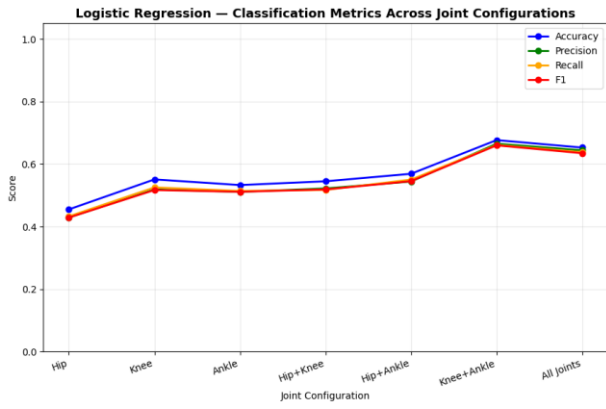


Figure 6. Accuracy, precision, recall, and F1 score for LR across joint configurations.

3.3 Confusion matrix analysis

Confusion matrices were examined for representative best-performing single-joint, two-joint, and three-joint configurations. The best single-joint configuration was knee-only KNN (Figure 7). Although many samples were correctly classified, several apparent equinus and jump gait samples were confused. Crouch gait was more readily identified, consistent with its sustained knee flexion during stance. Knee kinematics alone were insufficient to separate all four patterns reliably.

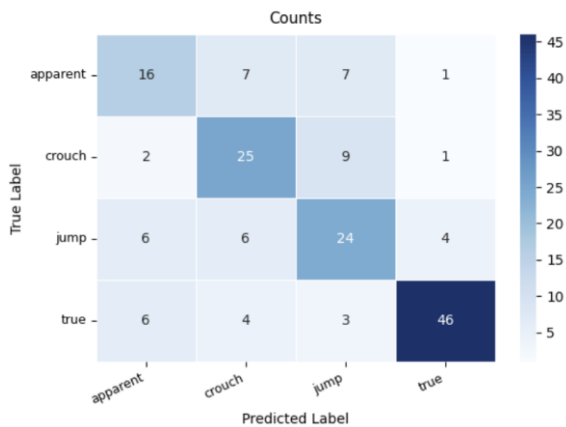


Figure 7. Confusion matrix for the best single-joint configuration, knee-only KNN.

Adding ankle motion to knee improved the confusion matrix (Figure 8). Correct predictions increased in several groups, and fewer apparent equinus samples were classified as crouch gait. Some confusion between true equinus and jump gait remained, potentially reflecting similarities during parts of the gait cycle.

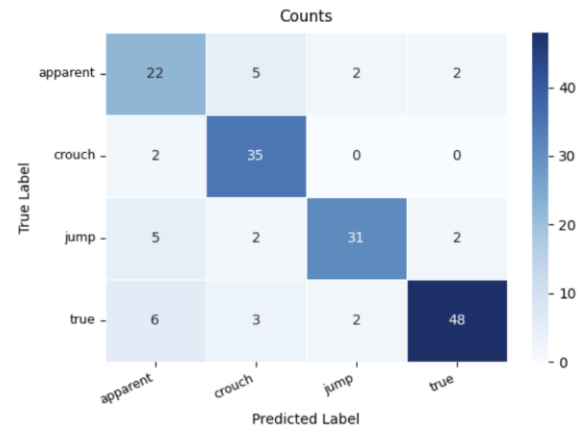


Figure 8. Confusion matrix for the best two-joint configuration, knee-and-ankle KNN.

The three-joint configuration produced the strongest overall pattern of correct classification (Figure 9). Crouch gait recall was high, and apparent equinus was identified with reasonable accuracy. True equinus also had more correct predictions, although some confusion with jump gait remained.

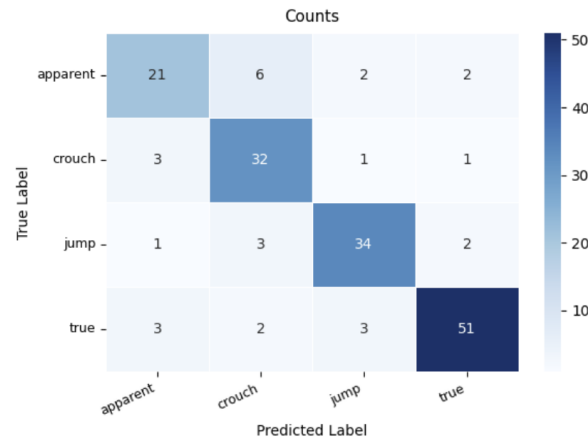


Figure 9. Confusion matrix for the best three-joint configuration, hip-knee-ankle KNN.

4 Discussion and conclusions

We compared four supervised classifiers for the four Rodda gait patterns using single-joint, two-joint, and three-joint sagittal kinematics. The primary aim was to assess how the hip, knee, and ankle contribute to distinguishing these classes.

No individual joint provided the same discrimination as the combined feature sets. The knee was the most informative single joint, whereas hip kinematics alone were less informative. This ranking agreed with the visual patterns in the mean trajectories and exploratory PCA plots.

Combining two joints improved the evaluation metrics, particularly for the knee-and-ankle configuration. Including all three joints produced the best KNN accuracy (0.8263) and F1 score (0.8122). These results are consistent with the coordinated, multijoint nature of the Rodda gait patterns; no formal significance test of the differences between configurations was reported.

KNN performed well in most configurations and had the highest three-joint test accuracy. Its performance may reflect nonlinear structure in the kinematic data. SVM was competitive, while LR generally performed worse. The observed model ranking is compatible with complex decision boundaries in this feature space, although it does not establish a unique underlying mechanism.

The confusion matrices showed persistent confusion between true equinus and jump gait in some configurations. Crouch gait was comparatively easier to identify, plausibly because its sustained knee flexion creates a more distinctive sagittal profile.

Overall, the joint-specific analysis provides both a practical comparison of classifiers and a biomechanical view of the information contributed by different joints. It may inform the development of decision-support tools that supplement visual clinical gait assessment.

This study has several limitations. It used only sagittal-plane joint angles; kinetic information and frontal- or transverse-plane kinematics were not included. Although cross-validation supported model selection, validation on an independent cohort would provide stronger evidence of generalizability. Future work could identify informative phases of the gait cycle with feature selection or importance analyses and compare additional machine learning approaches.

In summary, combining sagittal kinematics from the hip, knee, and ankle improved classification of the four Rodda patterns in this dataset and provides a basis for further evaluation in clinical gait analysis.

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